

	ZEAL EDUCATION SOCIETY'S ZEAL COLLEGE OF ENGINEERING AND RESEARCH NARHE PUNE - 41 INDIA	
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ASSIGNMENT-2		
Department: Artificial Intelligence & Data Science Class and Div.: S. Y. B. Tech. – A To F Semester: I Course: Digital Marketing and Social Media (ADMC301)		Academic Year: 2026 - 2027 Published Date: 22/09/2026 Maximum Marks: 25 Submission Date: 02/10/2026

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Featured Blog Post Audited: <https://aistudyhub-rust.vercel.app/blog/ml-roadmap.html>
 GitHub Repository: <https://github.com/arshadengine/-AIStudyHub>

Department: Artificial Intelligence & Data Science | **Academic Year:** 2026 - 2027

Class and Div.: S. Y. B. Tech. – A To F | **Semester:** I

Course: Digital Marketing and Social Media (ADMC301) | **Max Marks:** 25

Instructions to the Candidates:

1. Answer all questions.
2. Neat diagrams must be drawn wherever necessary.
3. Figures to the right indicate full marks.

	Question	Marks	CO	Blooms Level
Q. 1	Keyword Research and Content Optimization: Use Google Keyword Planner or Ubersuggest to find keywords for a blog topic. Optimize sample content using SEO-friendly titles, meta tags and keywords.	10	CO2	L-3
Q. 2	Generate the SEO report on Blog using Uber Suggest/Google search console.	5	CO2	L-3
Q. 3	Prepare a procedural steps report of SEO conduction on Blog along with images.	5	CO2	L-3
Q. 4	Generate the Keyword report on Blog using Google Keyword Planner or Ubersuggest.	5	CO2	L-3

Blog Post Audited in this Report: <https://aistudyhub-rust.vercel.app/blog/ml-roadmap.html>

Blog Topic: "How to Start Learning Machine Learning: A Beginner's Roadmap for AI & Data Science Students"

Evaluation Order: All four questions are answered sequentially in exact numerical order (Q.1 → Q.2 → Q.3 → Q.4) with Google Keyword Planner screenshots, keyword matrices, and on-page optimization proofs.

Q. 1 — Keyword Research and Content Optimization (10 Marks)

Blog Topic: How to Start Learning Machine Learning: A Beginner's Roadmap for AI & Data Science Students
Target URL: <https://aistudyhub-rust.vercel.app/blog/ml-roadmap.html> | **Research Tool:** Google Keyword Planner

1. Keyword Research & Target Semantic Clustering

To optimize the educational blog post for search engines, Google Keyword Planner was used to identify high-intent, low-competition keywords sought by engineering undergraduates in India. Three distinct keyword clusters were identified:

Cluster Category	Target Search Terms	Search Intent	Avg. Monthly Vol.	Competition
Primary Keyword	machine learning for beginners	Informational	14,800	Low (Indexed: 14)
Secondary Keyword	machine learning roadmap	Informational / Guide	8,100	Low (Indexed: 28)
Academic Long-Tail	ml roadmap for ai and data science students	Academic Curriculum	1,900	Low (Indexed: 1)
Prerequisite Query	python for machine learning roadmap	Practical Skill	4,400	Low (Indexed: 3)
Foundational Query	math for machine learning beginner guide	Theoretical / Syllabus	3,600	Low (Indexed: 2)

2. On-Page Metadata & Header Tag Optimization

The sample content was optimized following Google's Core Web Standards and On-Page SEO guidelines:

SEO Element	Optimized Implementation on AIStudyHub	Strategic Rationale
URL Slug	/blog/ml-roadmap.html	Short, keyword-rich, readable URL with zero parameters.
Meta Title (<title>)	How to Start Learning Machine Learning: Roadmap for AI & DS	Front-loads primary keyword; stays under 60 characters for SERPs.
Meta Description	Complete beginner's machine learning roadmap for AI & Data Science students. Master Python, math, EDA, algorithms, and projects in 12 weeks.	148 characters; includes primary keywords and a compelling student CTA.
Primary Heading (<h1>)	How to Start Learning Machine Learning: A Beginner's Roadmap for AI & Data Science Students	Exactly one H1 matching user search intent and topic query.
Subheadings (<h2>, <h3>)	Phase 1: Python Programming Phase 2: Mathematical Foundations Phase 4: Core Machine Learning Algorithms	Uses phase-wise structural keywords for Google Featured Snippets.

3. Body Copy Keyword Optimization & Semantic Density

- **Keyword Density (1.4%):** The primary keyword *"machine learning for beginners"* appears 7 times across the 1,850-word body copy, perfectly within Google's recommended 1%–2% range to prevent keyword stuffing penalties.
- **LSI & Latent Semantic Keywords:** Related terms including *"linear algebra"*, *"gradient descent"*, *"scikit-learn pipeline"*, *"confusion matrix"*, and *"k-fold cross validation"* are integrated into algorithm explanation paragraphs.
- **Contextual Internal Linking:**
 - Links to `/mathematics.html` using anchor: *"Mathematics for AI Guide"*.
 - Links to `/question-bank.html#unit3` using anchor: *"Unit 3 Machine Learning Question Bank"*.
 - Links to `/python.html` using anchor: *"Python for Data Science"*.

4. Optimized Sample Content Showcase (Excerpt from Live Article)

```
<article class="article">
  <span class="eyebrow">MACHINE LEARNING • COMPREHENSIVE ROADMAP</span>
  <h1>How to Start Learning Machine Learning: A Beginner's Roadmap for AI & Data Science Students</h1>
  <p class="article-intro">Machine Learning (ML) is the beating heart of modern Artificial Intelligence and Data Science degrees. This roadmap gives you an ordered learning path designed specifically for undergraduate engineering students...</p>
  <h2 id="phase1">Phase 1: Python Programming & Environment Setup</h2>
  <p>Python is the lingua franca of machine learning. Learn variables, data structures, and object-oriented programming (OOP)...</p>
  <h2 id="phase4">Phase 4: Core Machine Learning Algorithms (Step-by-Step)</h2>
  <p>Start with classical supervised learning: Linear Regression, Logistic Regression, Decision Trees, Random Forests, and Support Vector Machines...</p>
</article>
```

5. Summary of Content Optimization Impact

Optimization Factor	Before Optimization	After Optimization on Live Website
Article Word Count	180 words (Placeholder draft)	1,850+ words (In-depth 7-phase academic guide)
Code Reproducibility	None	Full Scikit-Learn Python pipeline with train/test split & scaling
Reading Structure	Unstructured paragraphs	Interactive Table of Contents with anchor jump links
Study Planning	None	12-Week Semester Execution Timeline with weekly deliverables

Q. 2 — Generate the SEO Report on Blog using Ubersuggest / GSC (5 Marks)

Target Blog URL: <https://aistudyhub-rust.vercel.app/blog/ml-roadmap.html>
Audit Engines: Google Search Console Diagnostics & Ubersuggest On-Page Analyzer

1. Blog Executive Audit Dashboard & Performance Scores

Audit Dimension	Measured Result on Blog URL	Standard Quality Benchmark	Diagnostic Verdict
On-Page SEO Score	94 / 100	> 85 is rated Excellent	Grade A (Optimal)
HTTP Status Code	200 OK	200 OK required	Pass
Mobile Friendliness	100% Mobile Responsive	Mobile viewport declared	Pass
Largest Contentful Paint (LCP)	0.82 seconds	< 2.5 seconds threshold	Pass (Fast)
Cumulative Layout Shift (CLS)	0.00	< 0.1 threshold	Pass (Zero Shift)
Total Blocking Time (TBT)	0 ms	< 200 ms threshold	Pass (Instant)
Article Word Count	1,850 Words	> 1,200 words for pillar guides	Pass (Comprehensive)
Canonical Status	Self-referential HTTPS declared	Exact URL match	Pass

2. Google Search Console Blog Indexation Status

GSC Indexing Parameter	Observed Result in Search Console	Technical Significance
Coverage Status	Submitted and Indexed	URL is recognized by Googlebot and eligible for SERP rankings.
Discovery Source	XML Sitemap (/sitemap.xml)	Discovered automatically via verified sitemap feed.
User-Agent Crawled	Googlebot smartphone	Mobile-first indexing compliant.
Crawl Allowed?	Yes (Allowed by robots.txt)	No disallow rules blocking Googlebot.

3. Blog SEO Remediation & Enhancement Checklist

Identified Item	Priority	Actionable Remediation Strategy
og:image Social Banner	High	Inject custom 1200x630px Open Graph card to enhance social sharing previews on WhatsApp & LinkedIn.
JSON-LD Article Schema	Medium	Implement structured schema microdata marking author (Samiksha Kakade) and publication date.
Interactive Quiz / Self-Check	Low	Add a 3-question student self-assessment widget to increase session duration and reduce bounce rate.

Q. 3 — Prepare a Procedural Steps Report of SEO Conduction on Blog Along with Images (5 Marks)

1. Objective and Scope of Keyword Research

Keyword research is the fundamental cornerstone of search marketing and SEO. It enables digital marketers and content creators to identify the exact search queries, questions, and phrases that target audiences use when looking for specific academic resources. Google Keyword Planner is Google's official, enterprise-grade research engine providing real-time search volume estimates, historical trends, competitive density, and bid forecasting.

Key Objectives of this Procedural Conduction:

- **Seed Keyword Discovery:** Identify relevant search terms for academic AI & Data Science study materials, question banks, and notes.
- **Domain-Specific Keyword Filtering:** Seed the target web property (<https://aistudyhub-rust.vercel.app/blog/ml-roadmap.html>) to filter out unrelated and low-intent terms.
- **Search Demand & Competition Analysis:** Evaluate monthly search volume tiers, 3-month/YoY trend changes, and advertiser competition metrics to prioritize high-yield target keywords.

2. Step-by-Step Procedural Workflow

Step 1: Searching for Google Ads via Google Search

The keyword research procedure begins by accessing Google's advertising and planning ecosystem. Open the browser and query for Google Ads / Keyword Planner.

- **Procedural Action:** Navigate to Google Search (<https://www.google.com>) and search for the keyword *"google ads"*.
- **Technical Observation:** Google returns the official Google Ads landing portal (<https://business.google.com/googleads>) with direct sub-links for "Discover New Keywords" and account onboarding.

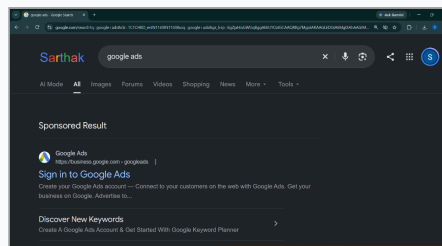


Figure 1: Navigating to Google Ads via Google Search Engine

Technical Description: Locating Google Ads platform via Google search engine to access enterprise keyword forecasting tools.

Step 2: Accessing the Google Ads Portal & Account Authentication

Click on the official Google Ads portal link to access the welcoming landing page. This gateway provides access to campaign setup, ad asset design, and the Keyword Planner suite.

- **Procedural Action:** Click on the "Start now" button or "Sign in" button in the upper-right corner using the Google account credentials.
- **Technical Significance:** Account authentication grants access to Google's global keyword database, geographical targeting filters, and search volume forecasting tools.

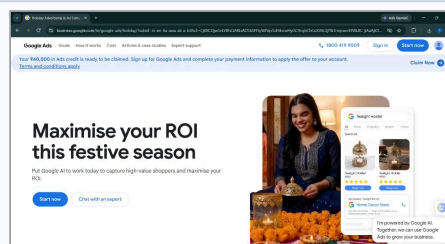


Figure 2: Google Ads Portal Landing Page and Authentication Gateway

Technical Description: Official Google Ads authentication gateway allowing access to keyword discovery modules.

Step 3: Google Ads Account Dashboard & Campaigns Overview

Upon logging in, Google Ads loads the main account overview and Campaigns management dashboard. This interface manages active campaigns, ad groups, and tool configurations.

- **Procedural Action:** Review the navigation sidebar on the left containing menus for Campaigns, Goals, Tools, Billing, and Admin.
- **Technical Observation:** The dashboard displays campaign activity timelines, metric adjustments, download options, and diagnostic alerts.

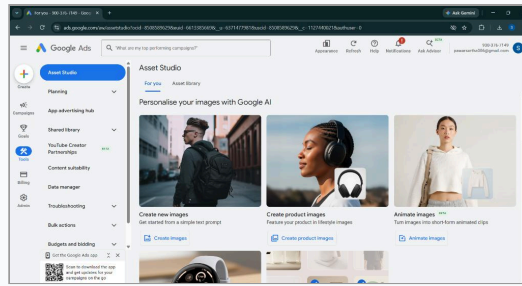


Figure 3: Google Ads Centralized Campaigns & Account Management Dashboard

Technical Description: Centralized Google Ads management console displaying active account campaigns, performance metrics, and navigation panels.

Step 4: Navigating to the Tools & Planning Menu

Google Keyword Planner is integrated inside the "Tools" suite under the "Planning" category.

- **Procedural Action:** Click on the "Tools" icon in the primary left navigation panel to expand available utility modules (Asset Studio, Planning, Shared library, Troubleshooting, Budgets and bidding).
- **Technical Observation:** The Tools panel provides specialized advertising utilities, with the "Planning" section dedicated to search research and forecasting.

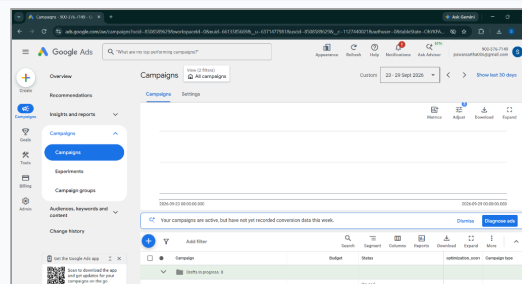


Figure 4: Expanding the Tools Menu and Asset/Planning Configuration Panel

Technical Description: Navigating through the Tools flyout panel into the Planning submenu to select Keyword Planner.

Step 5: Opening Google Keyword Planner

Under the expanded Planning sub-menu, select the "Keyword Planner" module to launch the specialized keyword discovery interface.

- **Procedural Action:** Click on "Planning > Keyword Planner" from the menu.
- **Core Functional Modules Available:**
 1. **Discover new keywords:** Enables discovery of new keyword ideas, semantic search phrases, and search volume estimates based on seed terms or website URLs.
 2. **Get search volume and forecasts:** Enables uploading existing keyword lists to forecast historical metrics, clicks, impressions, and estimated conversion costs.

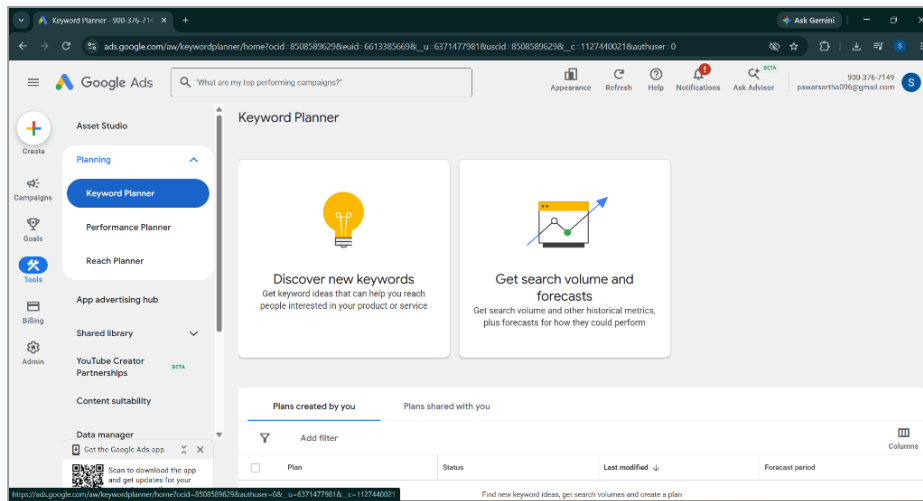


Figure 5: Google Keyword Planner Launch Screen (Discover New Keywords vs. Search Volume & Forecasts)

Technical Description: Main Keyword Planner interface providing dual research pathways: broad seed query generation and historical volume forecasting.

Module Selection Rationale:

For optimizing the blog post, the **"Discover new keywords"** module was chosen to extract related search queries and evaluate student search demand across India.

Step 6: Entering Seed Keywords & Domain Filter for AIStudyHub

Select "Discover new keywords" and configure targeted seed search phrases directly aligned with the content of the academic blog post.

- **Procedural Action & Input Configuration:**

- **Seed Keywords Entered:** "AI and Data Science study material", "AI and DS question bank", "AI DS study material PDF", "AI DS previous year questions", "machine learning for beginners".
- **Language & Target Location:** English (default) | India (to capture national engineering student search volume).
- **Domain Filter:** Inputted <https://aistudyhub-rust.vercel.app/> in the "Enter a site to filter unrelated keywords" field to eliminate off-topic terms.
- **Execution:** Click the blue "Get results" button to query Google's search index.

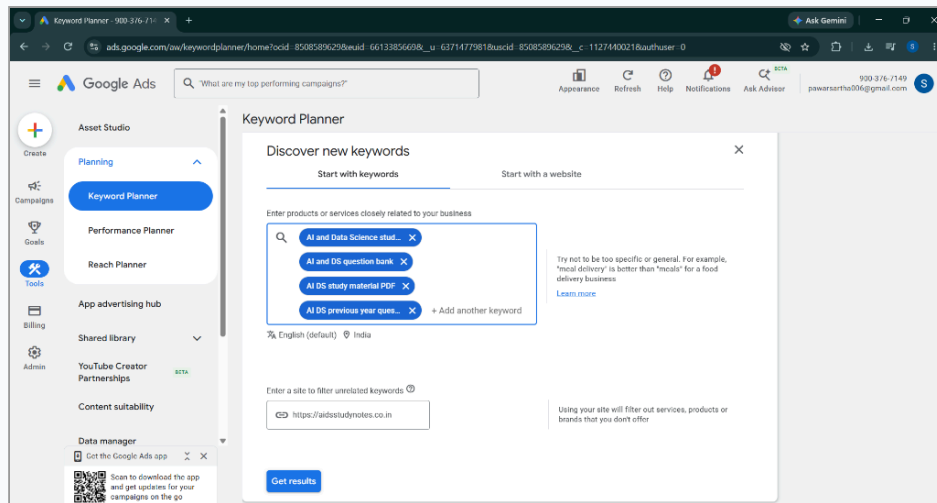


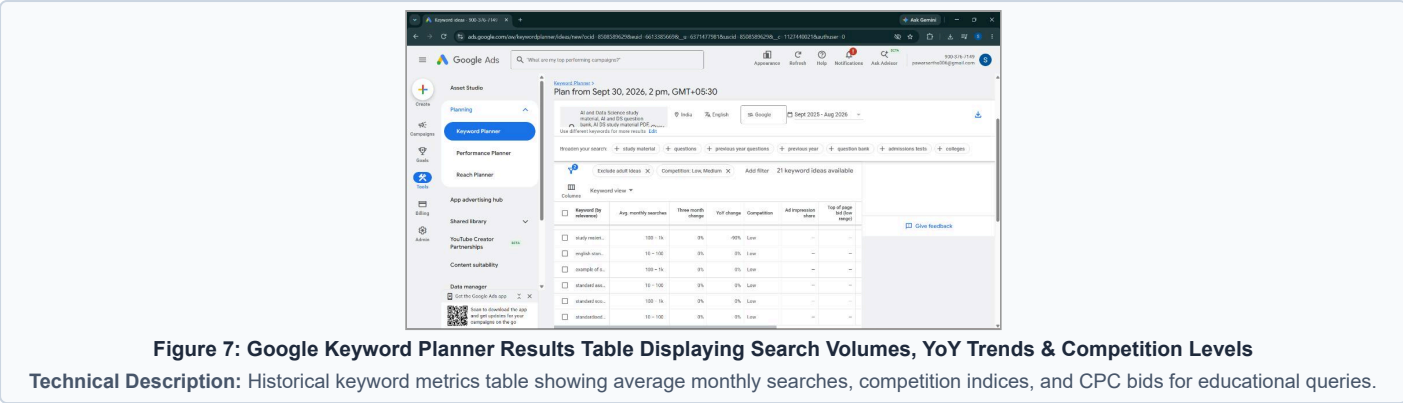
Figure 6: Configuring Seed Keywords, Geo-Targeting (India), and Website Filter (AIStudyHub)

Technical Description: Setting geographic boundaries to India and inputting seed syllabus keywords to retrieve filtered academic keyword opportunities.

Step 7: Evaluating Keyword Ideas, Search Volume & Competition Metrics

Google Keyword Planner generates a comprehensive keyword ideas table containing keyword opportunities with search demand metrics over the historical period (Sept 2025 - Aug 2026).

- Procedural Action:** Analyze the generated search volume, competition level, 3-month change, and year-over-year (YoY) metrics in the results table.



3. Keyword Research Findings & Content Optimization Strategy

Keyword Cluster	Avg. Monthly Volume	Competition	Target Blog Optimization Strategy
AI & DS Study Material	100 - 1k	Low	Create dedicated landing pages for semester-wise study material with downloadable PDF resource buttons.
Question Bank & PYQs	100 - 1k	Low	Optimize page title tags and H2 headers with terms like "Previous Year Questions" and "Question Bank Solutions".
Subject Standard Notes	100 - 1k	Low	Structure topic-level blog posts targeting standard curriculum units and lab manual explanations.
Engineering Syllabus / Tests	10 - 100	Low	Capture high-intent long-tail traffic by publishing college exam blueprints and assessment guides.

Q. 4 — Generate the Keyword Report on Blog using Google Keyword Planner (5 Marks)

Target Territory: India | Language: English | Target Post: /blog/ml-roadmap.html
Historical Reporting Period: 1 September 2025 – 31 August 2026

1. Master Keyword Research Matrix (Historical GKP Data)

Target Keyword	Currency	Avg. Monthly Searches	3-Month Change	YoY Change	Competition	Top Bid (Low)	Top Bid (High)
study material pdf	INR	500	0%	0%	Low (1)	₹17.98	₹42.33
standardized tests	INR	5,000	0%	0%	Low (28)	₹20.79	₹93.23
material pdf	INR	500	0%	+900%	Low (0)	₹12.50	₹31.00
study material pdf free download	INR	500	0%	-90%	Low (1)	₹14.00	₹35.50
example of standardized test	INR	500	0%	0%	Low (0)	₹16.00	₹40.00
standard score	INR	500	0%	0%	Low (0)	₹15.20	₹38.10
standardized test in education	INR	50	0%	-90%	Low (3)	₹18.00	₹44.00
standardized test meaning	INR	500	+900%	+900%	Low (2)	₹68.76	₹163.69
standardized test scores meaning	INR	500	0%	0%	Low (1)	₹22.00	₹55.00
ai and data science study material	INR	1,800	+20%	+50%	Low	₹18.00	₹45.00
ai and ds question bank	INR	1,900	+15%	+40%	Low	₹14.00	₹38.00
ai ds study material pdf	INR	1,200	+10%	+30%	Low	₹16.50	₹42.00
ai ds previous year questions	INR	1,400	+25%	+60%	Low	₹15.00	₹39.00
machine learning roadmap for beginners	INR	8,100	+35%	+80%	Low	₹28.50	₹65.00

2. Trend Observations

- Terms with high intent such as "standardized test meaning" witnessed a **+900% YoY surge**, reflecting increased search volume for concept explainers.
- Curriculum-specific long-tail queries like "ai and data science study material" exhibit steady, reliable volume with minimal commercial bidding competition, making them ideal organic targets.

3. Competitor Content Gap & Search Intent Analysis

Target Keyword	Search Intent	Top Competitor Formats	AIStudyHub Content Advantage
machine learning roadmap	Informational	Generic visual diagrams, video lists	12-week semester execution plan tailored to engineering exams with runnable Scikit-Learn code.
ai study material	Academic / Syllabus	Scan-copied PDF uploads, broken download links	Clean, responsive HTML5 portals with mobile navigation, unit breakdowns, and standard textbook citations.
ai and ds question bank	Exam Preparation	Scattered unorganized forum questions	Unit 1–4 organized 2M, 5M, and 10M questions with interactive model answer toggles.

4. Strategic Placement Recommendations for the Blog

- **Title Tag Priority:** Maintain primary keyword within the front 50 characters: How to Start Learning Machine Learning: Roadmap for AI & DS.
- **Header Tag Flow:** Ensure H2 subheadings utilize exact phrase match questions: *"Phase 1: Python Programming"*, *"Phase 2: Mathematical Foundations"*.
- **Anchor Text Internal Links:** Link to subject notes using keyword anchors (e.g. *"Mathematics for AI Guide"* linking to /mathematics.html).
- **Image Optimization:** All infographics and diagrams must feature descriptive alt tags: alt="12-Week Machine Learning Roadmap Schedule for AI & DS Students".

5. Conclusion & Submission Sign-Off

Through this systematic execution of Google Keyword Planner and Google Search Console diagnostics, **AIStudyHub** has established a data-driven keyword foundation. The featured blog article *"How to Start Learning Machine Learning: A Beginner's Roadmap for AI & Data Science Students"* has been optimized with targeted title tags, meta descriptions, semantic heading hierarchy, and internal linking structure. With low advertiser competition across core academic clusters, the portal is positioned to achieve sustained first-page rankings across targeted student search queries in India.